EPSR-test-ex102 Coarse graining with sphere parameters and sizefactor > 1

Program(s): epsr | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

What this case tests

Exercises the coarse-graining option of the `epsr` engine of EPSR [1,2], on when the first value of `qinscalefc` is > 1: per-molecule size factors `scalefacmol` (`.ato` molecule headers) and per-type sphere parameters `natomsph radminsph radmaxsph cgmolfac` define form factors and scale the reference-potential distances of molecule pairs [3]. This case sets non-default sphere parameters for every atom type and combines coarse graining with a declining box `sizefactor`.

Code exercised

Fortran (original, fortran_src/):

- `epsr_standalone_rev3.f`: `runepsr17` (`qinscalefc`), `inita` (coarse-graining block: bins `assignatomscalefac`, `sphereformfactor`, `intraformfactor`, `intraformfactorself`), `gaussianqr2`
- `molfit.f`: `en_mole_pair`, `en_change`, `en_atom_inter_scale` (pair scale factor `scalefac`)
- `ato_manipulation_rev4.f`: `atogen` (scaled `potex`, `wavinter`); `ato_read_and_save.f`: `minitc`, `moloutc`

Rust (port, rust/epsr-core/src/):

- `coarsegrain.rs` (new): `State::inita_coarsegrain`, `gaussianqr2`, `atogen_accumulate`, `atogen_normalise`
- `epsr.rs`: `Epsr::setup` (`qinscalefc`); `structure.rs`: `inita`, `atogen`
- `energy.rs`: `en_mole_pair`, `en_change`, `en_atom_inter_scale`; `ato.rs`: `minitc`, `moloutc`
- `epsr.rs`: `Epsr::iterate` (`sizefactor` decline)

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å^3). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [4] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [5]. Coarse graining represents atoms by uniform spheres (form factors from `gaussianqr2` of the inner and outer radius); a molecule pair scales the reference-potential distance by `scalefac` = `0.5*(scalefacmol(i)+scalefacmol(j))`. Here DMSO `scalefacmol` runs from 1.00 to 1.40 in steps of 0.05 (water 1.0); the type columns are S `-0.08 0.3 1.9 2.0` (negative `natomsph` = a density, converted with `cgmolfac`), C `1.0 0.5 2.0 2.0`, O `1.0 0.0 1.5 2.0`, H `1.0 0.0 0.0 2.0`, O1 `-0.05 0.0 1.7 1.0`, H1 `1.0 0.2 0.6 0.5` (`cgmolfac` < 1 is reset to 1 on reading).

Inputs

Generated from EPSR-test by `scripts/make_cg.py <folder> <mode>` (edits `.EPSR.inp` keywords, the `.ato` `scalefacmol` words and type columns). Mode for this case: `.EPSR.inp` `qinscalefc 3.0 0.5`, `ntimes 3`, `sizefactor 1.2 0.9 1.0E+06`; `.ato` as above; `.pcof` with the `rminex` values of EPSR-test-ex1.

How the accuracy was verified

`commands.txt` (1 program call, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 2 passes.
The huge threshold makes the size factor decline every iteration, 1.2 -> 1.08 -> 1.0 (clamped), which also stops intramolecular rotations while it is > 1. The three form-factor files are compared byte for byte with the standard outputs.

Notes

Single-precision quirk reproduced: int((1.4-1.0)/0.2)+1 gives 2 (not 3) bins for the DMSO types. Fortran: form factors and `wavinter` are computed but never used for S(Q) (the coarse-graining code in `gensq` is commented out).

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.